

SEDAR 49 Gulf of Mexico Data Limited Species

Data Workshop

LIFE HISTORY RESEARCH RECOMMENDATIONS

Red Drum

The SEDAR 49 Gulf of Mexico data-limited stock assessment represents the initial attempt at assessing Gulf of Mexico Red Drum since the federal harvest moratorium. A comprehensive review of the literature, as well as inclusion of the most recent datasets available, provided the most up to date life history information possible (Table 2.12.1, 2.12.4). Through this review of the literature, it is apparent that GOM Red Drum remain a data-limited species. Below we provide the following research recommendations:

1. Increase offshore sampling across the entire GOM, especially at the individual school level, for biological samples (e.g., meristics, otoliths, reproductive tissues, fin clips). We recommend purse seine as the least size-selective sampling gear for this species in offshore waters.
2. Consensus and consistency is needed in assigning calendar age, calculating fractional ages and recording edge type across the GOM to ensure the age data collected are comparable between studies.
3. A concerted effort should be made to identify and record reproductive phase for oocyte development, both macroscopically and histologically. This is particularly true given that the most recent reproductive estimates are greater than 20 years old. Improved quantification (e.g., binary logistic regression) is needed for better point estimates of size and age at 50% and 95% maturity.
4. Collection of tissues (e.g., fin clips) is a low-cost and easy-to-archive means to ensure future studies examining stock delineation, site fidelity, effective population size, etc. for this species are possible.

Lane Snapper

A primary open question in the life history analyses is how the recreational fishery has impacted the stock since the early 1990's. There are no data available to make inferences about how age frequency in the fishery and stock may have changed over the time series.

Primary research needs identified by the team included the following. These are listed below in order of priority based on perceived priority:

1. Increase the precision (by increasing sample size and thorough validation) of estimates of length-at-age and maturity-at-age to provide rigorous estimates. This would require an increase in dockside and at-sea sampling for biostatistical information, especially the collection of otoliths and reproductive tissue.

2. Design random sampling protocol for NMFS Pascagoula's groundfish and small pelagic surveys to collect length- and age-composition of Lane Snapper encountered by these surveys.
3. Perform a survey of the genetic structure of the stock to more precisely understand spatial stock structure, in particular the potential for hybridization with other Lutjanids.

Wenchman

Due to the limited sampling of life history parameters (two months of data in a single year), more research is needed for all life history aspects of Wenchman. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following:

1. Increase dockside and at-sea sampling for biological samples (age structures, reproductive tissues, and genetic material).
2. An aging study that includes validation with increased sample sizes.
3. Design a random sampling protocol for NMFS Pascagoula groundfish and small pelagic surveys.
4. Collect reproductive maturity estimates.

Yellowmouth Grouper

Additional research is needed to obtain more recent estimates of all life history parameters for Yellowmouth Grouper. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following:

1. Increase in dockside and at-sea sampling for biological samples (age structures, reproductive tissues, and genetic material) for the GOM.
2. Conduct an updated age and growth study for GOM samples, including a validation study based on radiochemical dating.
3. Conduct an updated reproductive study for the GOM to examine not only maturity but the size and age of transition.

Snowy Grouper

Additional research is needed to obtain more recent estimates of all life history parameters for Snowy Grouper in the GOM. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following:

1. Increase in dockside and at-sea sampling for biological samples (age structures, reproductive tissues, and genetic material) for the GOM.

2. Conduct an updated age and growth study for GOM samples, which also includes a more extensive validation study based on radiochemical dating (see Harris 2005).
3. An increase in dockside and other sampling programs to complete a more comprehensive and an updated reproductive study for GOM to examine not only maturity but size and age of transition.

Speckled Hind

Additional research is needed to obtain estimates of all life history parameters for Speckled Hind in the northern GOM. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following:

1. Increase in dockside and at-sea sampling for biological samples (age structures, reproductive tissues, and genetic material) for the GOM.
2. Conduct an updated age and growth study for GOM samples, using the new criteria of counting narrower groups of translucent and opaque band increments on the dorsal side of the otolith (as described in Andrews et al. 2013).
3. An increase in dockside and other sampling programs to complete a more comprehensive and an updated reproductive study for the GOM to examine not only maturity but size and age of transition.

Lesser Amberjack

Additional research is needed to obtain estimates of all life history parameters for Lesser Amberjack in the GOM. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following.

1. Increase in dockside and at-sea sampling for biological samples including age structures, reproductive tissues, and genetic material.
2. While age has been attempted, finding an appropriate aging methodology that includes a way to validate age using multiple hard structures is suggested.
3. Further research is needed for natural mortality estimates.
4. Need for reproductive tissue to examine maturity.

Almaco Jack

Additional research is needed to obtain estimates of all life history parameters for Lesser Amberjack in the GOM. This includes aging, reproduction and maturity, and estimation of growth parameters.

Primary research needs identified by the LHWG included the following.

1. Increase in dockside and at-sea sampling for biological samples including age structures, reproductive tissues, and genetic material.
2. While age has been attempted, finding an appropriate aging methodology that includes a way to validate age using multiple hard structures is suggested.
3. Further research is needed for natural mortality estimates.
4. Need for reproductive tissue to examine maturity.

COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

Further development of methods for calculating overall uncertainty when summing total removals from commercial, recreational, and other fisheries (e.g., shrimp and other trawl fisheries). Methods should account for differences in programs; e.g., some programs provide CVs while others produce ranges of uncertainty based upon expert opinion.

Develop more robust estimates of discard mortality for all SEDAR 49 species from each sector of the commercial fishery.

Develop methods to more appropriately estimate uncertainty of discard estimates from each sector of the commercial fishery.

Red Drum

Develop data collection methods to enable investigation of the magnitude of bycatch in the Gulf of Mexico menhaden fishery for Red Drum. Investigate the impact of menhaden fishery bycatch on stock assessments.

Lane Snapper

Develop appropriate sampling methods to determine the size composition of Lane Snapper caught as bycatch in Gulf of Mexico shrimp fisheries.

Wenchman

During the Data Workshop, a northern Gulf of Mexico finfish trawl fishery (likely targeting Butterfish) was identified as being the primary commercial fishery for Wenchman. That fishery was recommended as the representative fleet for Wenchman. Further investigation of that finfish trawl fishery is recommended. Data sources useful for accurately determining targeting, effort, and landings of the fishery should be identified.

Develop appropriate sampling methods to determine the size composition of Wenchman caught as bycatch in Gulf of Mexico shrimp fisheries.

Yellowmouth Grouper

Develop genetic markers for species identification and determine the frequency of misidentification of Yellowmouth Grouper.

Use port samplers to determine the frequency of Yellowmouth Grouper misidentification or misreporting.

Snowy Grouper

No research recommendations were suggested for Snowy Grouper.

Speckled Hind

No research recommendations were suggested for Speckled Hind.

Lesser Amberjack

Use port samplers to determine the frequency of Lesser Amberjack misidentification or misreporting.

Almaco Jack

Use port samplers to determine the frequency of Almaco Jack misidentification or misreporting.

RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

Red Drum

- Improve discard length and age data collection in the recreational fishery.
- Develop directed effort estimates.
- Investigate self-reported discards to determine if there is bias or misidentification in the data.
- Determine implications of gaps in the available recreational discard data.

Lane Snapper

- Improve discard length and age data collection in the recreational fishery.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.

Wenchman

- Improve discard length and age data collection in the recreational fishery.
- Determine whether species identification issues (not commonly known in the recreational fishery) affect reported landings/discards.
- Reliable estimates of discard mortality.

- Develop directed effort estimates.

Yellowmouth Grouper

- Improve discard length and age data collection in the recreational fishery.
- Determine whether species is underreported and the percentage of landings/discards underreported due to species misidentification as Scamp or Black Grouper.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.
 - Species that are not typically targeted (ex: Yellowmouth Grouper) may benefit from a higher-level directed effort estimate (ex: shallow water grouper effort), as they are frequently caught in conjunction with associated species.

Snowy Grouper

- Improve discard length and age data collection in the recreational fishery.
- Determine whether species is underreported and the percentage of landings/discards underreported due to species misidentification as Black Grouper or Warsaw Grouper.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.

Speckled Hind

- Improve discard length and age data collection in the recreational fishery.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.
- Investigate self-reported discards to determine if there is bias or misidentification in the data.
- Determine implications of gaps in the available recreational discard data.

Lesser Amberjack

- Improve discard length and age data collection in the recreational fishery.
- Determine effect of misreporting due to species misidentification as Banded Rudderfish or Greater Amberjack.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.

Almaco Jack

- Improve discard length and age data collection in the recreational fishery.
- Determine whether dead discards are underestimated in TX due to targeted bait fishery.
- Reliable estimates of discard mortality.
- Develop directed effort estimates.
 - In Texas there is a unique bait fishery which targets Almaco Jack. It was noted that b1 may be underestimated in Texas. It may be worth investigating the directed effort from this fishery.
- Investigate self-reported discards to determine if there is bias or misidentification in the data.
- Determine implications of gaps in the available recreational discard data

TOTAL REMOVALS RESEARCH RECOMMENDATIONS

See recommendations in Sections 3.6 and 4.6.

MEASURES OF FISHING EFFORT RESEARCH RECOMMENDATIONS

See recommendations in Sections 3.6 and 4.6.

INDICES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

Red Drum

Given the importance of Red Drum to the recreational fishing interests of the Gulf Coast States, it was surprising to find that a survey designed to comprehensively sample both the near shore and offshore portions of the Gulf of Mexico stock does not exist. It is recommended that discussions be initiated into expanding an existing survey or developing a new survey to sample and characterize the composition and relative abundance of the Gulf of Mexico Red Drum stock, especially in federally managed waters where little data are available.

Lane Snapper

No research recommendations were suggested for Lane Snapper.

Wenchman

The small pelagics survey used as the index of abundance for SEDAR 49 is no longer in operation. The deep-water sampling of this survey provided the only data on a largely otherwise un-surveyed portion of the Gulf of Mexico Wenchman stock. Additional resources need to be put forward to promote and expand deep-water sampling efforts in the Gulf for species like Wenchman and numerous other deep-water species.

Yellowmouth Grouper

Additional information about Yellowmouth Grouper distribution and habitat utilization is needed to determine if low counts in the reef fish video survey are due to low abundance or survey habitat mismatch.

Snowy Grouper

Surveys designed to better cover deep-water habitat are needed to adequately sample the Snowy Grouper stock as well as many other reef fish managed under the reef fish FMP.

Speckled Hind

Surveys designed to better cover deep-water habitat are needed to adequately sample the Speckled Hind stock as well as many other reef fish managed under the reef fish FMP.

Lesser Amberjack

Species identification issues are of paramount concern for Lesser Amberjack, especially when dealing with fishery-dependent data sources. Efforts should be undertaken to determine whether port sampling data can be used to estimate the rate at which species like Lesser Amberjack are misidentified on an annual basis. This information could be used to adjust fishery-dependent landings data, allowing them to be used to construct indices of relative abundance.

Almaco Jack

Species identification issues are of paramount concern for Almaco Jack, especially when dealing with fishery-dependent data sources. Efforts should be undertaken to determine whether port sampling data can be used to estimate the rate at which species like Almaco Jack are misidentified on an annual basis. This information could be used to adjust fishery-dependent landings data, allowing them to be used to construct indices of relative abundance.

LENGTH FREQUENCY RESEARCH RECOMMENDATIONS*Red Drum*

- Continue and expand fishery-independent collection efforts to collect length measurements at varying sizes, seasons or months, and locations, particularly for offshore Red Drum

Lane Snapper

- Continue and expand collection efforts to collect length measurements at varying locations, seasons or months
- Pursue statistical approaches to address sampling inconsistencies between random selection of small and large individuals in the SEAMAP groundfish survey, which could enable the use of length composition derived from the SEAMAP groundfish survey

Wenchman

- Continue and expand collection efforts to collect length measurements at varying locations, seasons or months
- Create sampling protocols to obtain lengths from NMFS Pascagoula small pelagic survey

Yellowmouth Grouper

- Expand collection efforts to collect genetic samples to ensure species identification along with length measurements at varying locations, seasons or months

Snowy Grouper

- Continue and expand collection efforts to collect length measurements at varying locations, seasons or months

Speckled Hind

- Continue and expand collection efforts to collect length measurements at varying locations, seasons or months

Lesser Amberjack

- Expand collection efforts to collect genetic samples to ensure species identification along with length measurements at varying locations, seasons or months

Almaco Jack

- Expand collection efforts to collect genetic samples to ensure species identification along with length measurements at varying locations, seasons or months

AGE FREQUENCY RESEARCH RECOMMENDATIONS*Red Drum*

- Develop common practices for aging, interpreting edge, assigning annual or co-hort age, and calculating fractional age (or biological age) for Red Drum across federal and state agencies
- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations, particularly for offshore fish

Lane Snapper

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001)

Wenchman

- Increase collection of age samples at varying sizes, seasons or months, and locations
- Determination of the reproductive season to assist in determining when growth increments are deposited
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001)

Yellowmouth Grouper

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001).

Snowy Grouper

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001).

Speckled Hind

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001).

Lesser Amberjack

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Improvement of methods for aging *Seriola* sp. due to the difficulty in interpreting annuli marks
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001)

Almaco Jack

- Expand collection efforts to collect age samples at varying sizes, seasons or months, and locations
- Improvement of methods for aging *Seriola* sp. due to the difficulty in interpreting annuli marks
- Validation of annual increments using methods such as tag and recapture, mark-recapture of chemically tagged fish, captive rearing from hatch, and radiochemical dating (Campana 2001).

Assessment Workshop

A number of research recommendations were identified throughout the SEDAR 49 stock evaluation and are described below. Research recommendations for improvements to input data, which were provided at the end of each relevant section in the Data Report, are also reiterated by the analysts.

1. Fine-tuning of the index-based and length-based methods reported herein to achieve target performance metrics (e.g. probability of not overfishing closest to 50% or the highest LTY).

2. Exploration of the cost or benefit of specifying an operating model incorrectly and how this influences method selection over a range of operating model input parameters.
3. Calculation and presentation of performance metrics in relation to the status quo rather than a reference method.
4. Simulation testing of the non-equilibrium mean length estimator and yield-per recruit approach to assess method performance in comparison to other available methods, as well as testing different assumptions inherent in the approach (e.g. whether to use a time series of recent total removals or the terminal year's total removals in catch recommendations).
5. Evaluation of the updated Hoenig equation (described in Then et al. 2014) for estimating natural mortality using maximum age. The updated equation tends to produce higher estimates of natural mortality, which can have important implications for applications such as the mean length estimator.
6. Region-specific estimates of correlation coefficients for growth parameters derived from growth curves specific to the Gulf of Mexico.
7. Investigation of more justifiable estimates of stock depletion such as through Productivity-Susceptibility Analysis (Cope et al. 2015).
8. Estimation of current stock abundance from tagging studies (e.g. Red Drum), which could be used in methods such as the Beddington and Kirkwood (2005) approach.
9. Identification of a reference period for catches for Red Drum.
10. Discussions regarding the appropriateness of the reference period selected for each species.
11. Evaluation of the appropriateness of target catch or index levels which could be used in conjunction with catch and index time series.
12. Evaluation of the appropriateness of target length levels which could be used in conjunction with catch and a length frequency series.
13. Incorporation of observation error into the application of index-based (I_{slope0} , $I_{target0}$) and length-based ($L_{target0}$, $L_{stepCC0}$) methods.
14. Future data-limited assessments should ensure that the reliability scores for data inputs are agreed upon at the conclusion of the Data Workshop to provide a more quantitative means of weighting methods for catch recommendations.

Within the modeling framework used in SEDAR 49, many limitations are acknowledged within the MSE approach. Pragmatically, results are a product of the specific conditions of the simulation, which are assumed to be as simplistic as possible but contain sufficient complexity to reflect the system in a representative way. Thus, additional considerations towards confirmation of the stock and fleet subclass components of the operating models explored in SEDAR 49 are warranted. In addition, no implementation error was considered in the current analysis which employed the DLMtool Version 3.2.1.

Recommendations for enhancing the practical use of the DLMtool from the analytical team.

1. Revisions of the DLMtool software to enhance the model functionality to allow multiple fishing fleets.
2. Revision of the DLMtool software to allow age varying natural mortality.
3. Allow for implementation error of the harvest control rule (e.g. catch recommendation overages) within the implementation model in the MSE.

Review Workshop

- Sea sampling programs to better quantify discards and discard mortality for all the eight species.
- The choice of reference time period for Tier 3A and Tier 3B stocks needs to be re-visited given the new information available and possible changes in the ecosystems.
- The operating model simulates the population dynamics of a given species conditional on the assumed depletion level which is usually unknown. Although the base case scenario for depletion level was developed for each species based on the best available information and a sensitivity analysis was conducted for alternative depletion levels, a reality check may be necessary to help simulate a fishery that realistically reflects the dynamics of fishery of interests. Reliable information on the fishery and population (e.g., temporal trend of fishing efforts, fishery-dependent and fishery-independent abundance indices and biological information such as age- and length compositions) needs to be collected to help define possible depletion level. These data can be used to tune the operating model parameterization to improve the fishery simulation realism by the operating models. Further, a number of surveys were considered at the DW but not all of them were deemed appropriate to inform a stock assessment. It is important to revisit the design of the surveys to ascertain whether changes could be made to get more value out of those surveys. The Review Panel also recommends that more time is spent to identify the methodology and indicators that are best for the type of exploitation and species we have. Trying to calculate MSY and other conventional metrics might not be the most

appropriate approach especially for species that are caught as bycatch. Similarly, collecting all the data that are needed to do a proper stock assessment is a very big task and it is important to identify some interim approaches such as using indicator species (to represent a complex of species) or maybe use the status of the targeted stock as a proxy for the status of the by-catch species.